

ClinicaFlow

AI Diagnostic Summary Report

Patient Overview

- **Patient Name:** Ashley Grahm
- **Age:** 20
- **Gender:** Female
- **Primary Complaint:** stomach pain and diarrhea

Red-Flag Alert

- **Severe hypoxemia** ($\text{SpO}_2 = 85\%$)
- **Leukocytosis** ($\text{WBC} = 10,570$) suggesting infection/inflammation
- **Orthostatic dizziness** indicating possible dehydration or early circulatory compromise
- **Hyperglycemia & new-onset diabetes** (fasting glucose = 141 mg/dL, $\text{HbA1c} = 7.1\%$)

Risk Stratification

- **Level:** High
- **Key Drivers:**
 - Low oxygen saturation ($\text{SpO}_2 85\%$)
 - Elevated WBC count (10,570)
 - Orthostatic dizziness/weakness
 - Hyperglycemia with HbA1c in diabetic range
 - Acute abdominal pain & diarrhea < 24 h after a possible foodborne exposure

Probable Diagnosis

- **Condition:** Acute infectious gastroenteritis with dehydration and secondary hypoxemia (possible early sepsis)
- **Confidence:** 78 %
- **Justification (Explainability Pack):**
 - The patient reports “**stomach pain and diarrhea**” that began “**this morning after pizza consumption.**”
 - Laboratory data show **WBC 10,570 (high)**, supporting an infectious/inflammatory process.
 - **$\text{SpO}_2 85\%$** together with orthostatic dizziness points to hypoxemia likely secondary to dehydration and/or early systemic response.
 - **Fasting glucose 141 mg/dL** and **$\text{HbA1c} 7.1\%$** reveal previously undiagnosed diabetes, a factor that can worsen infection severity.

• **Reasoning:** The combination of acute gastrointestinal symptoms, leukocytosis, hypoxemia, and metabolic derangements aligns with an infectious gastroenteritis complicated by volume depletion and early sepsis.

• **Supporting Evidence:**

- Abdominal pain and diarrhea started this morning after pizza consumption
- WBC 10,570 (high)
- SpO₂ 85 % (hypoxia)
- Orthostatic dizziness, weakness, inability to swallow comfortably
- Fasting glucose 141 mg/dL and HbA1c 7.1 % (new-onset diabetes)
- No documented fever, tachycardia, or hypertension

Differential Diagnoses

- **Foodborne bacterial gastroenteritis** (e.g., *Salmonella*, *Campylobacter*, *Staphylococcus aureus*) – recent pizza, acute onset, leukocytosis.
- **Early sepsis secondary to gastrointestinal infection** – hypoxemia, leukocytosis, dizziness, possible inadequate perfusion.
- **Dehydration with orthostatic hypotension** – dizziness on standing, low SpO₂, poor fluid intake.
- **Undiagnosed type 2 diabetes mellitus presenting with hyperglycemia and infection** – elevated fasting glucose and HbA1c.
- **COVID-19 or other viral respiratory infection presenting atypically** – low SpO₂ without respiratory complaints warrants exclusion.
- **Allergic reaction / atopic disease (high IgE)** – markedly elevated IgE, though no rash or airway symptoms reported.

Key Laboratory Findings

- **WBC Count:** 10,570 (High) – suggests infection/inflammation.
- **Fasting Blood Sugar:** 141 mg/dL (High) – indicates hyperglycemia.
- **HbA1c:** 7.10 % (High) – diagnostic of diabetes.
- **Vitamin B12:** < 148 pg/mL (Low) – deficiency.
- **Homocysteine:** 23.86 µmol/L (High) – cardiovascular risk marker.
- **IgE:** 492.30 IU/mL (High) – possible atopic/allergic predisposition.
- **25-OH Vitamin D:** 8.98 ng/mL (Low) – severe deficiency.

Recommended Plan

Required Investigations

Test	Purpose / What it Checks
CBC with differential	Confirm leukocytosis, assess anemia or other cell line changes
Comprehensive metabolic panel	Electrolytes, renal & hepatic function, glucose trends

Test	Purpose / What it Checks
Serum lactate	Detect tissue hypoperfusion / early sepsis
Arterial blood gas with pulse oximetry	Quantify hypoxemia, acid-base status
Blood cultures (2 sets)	Identify bacteremia/septic source
Stool culture, ova & parasites, C. difficile toxin PCR	Identify bacterial, parasitic, or C. diff infection
Rapid COVID-19 PCR or antigen test	Exclude viral respiratory cause of hypoxia
Chest X-ray	Evaluate for pneumonia or other pulmonary pathology
Abdominal ultrasound	Rule out appendicitis, intra-abdominal abscess, organomegaly
Urinalysis & urine culture	Assess dehydration, possible urinary infection
Serum ketones (β-hydroxybutyrate)	Detect ketoacidosis in new-onset diabetes
ECG (baseline)	Identify cardiac effects of electrolyte shifts or hypoxia
Serum vitamin B12, 25-OH vitamin D, homocysteine (repeat if needed)	Monitor deficiencies identified
Allergy panel or serum tryptase (if allergic reaction suspected)	Clarify high IgE significance

Suggested Medications / Treatments

- **IV isotonic crystalloid fluids** (e.g., normal saline 1–2 L bolus, then maintenance) – rehydration.
- **Oral rehydration solution** (if patient tolerates oral intake).
- **Antiemetic** – ondansetron 4 mg IV/PO q8h PRN.
- **Empiric broad-spectrum antibiotic** – ciprofloxacin 500 mg PO BID (if bacterial gastroenteritis strongly suspected or fever develops).
- **Supplemental oxygen** – titrate to maintain SpO₂ > 94 % (nasal cannula or mask).
- **Insulin sliding scale** – if blood glucose remains >180 mg/dL.
- **Vitamin B12 replacement** – cyanocobalamin 1000 µg IM weekly (after stabilization).
- **Vitamin D supplementation** – cholecalciferol 2000 IU daily (after acute issues resolved).

Medication disclaimer: **A qualified human doctor must make the final prescribing decision.**

Information Gaps *(if any)*

- Detailed allergy history (drug, food, environmental).
- Current medication/supplement list.
- Exact oral fluid intake and urine output over the past 24 h.
- Presence or absence of blood/mucus in stool.
- Any shortness of breath, chest discomfort, or cough that could explain hypoxia.
- Recent travel or known sick contacts beyond the pizza exposure.

Understanding Your Report — In Plain Language

This section explains your diagnostic report in everyday language. It is intended to help you understand what your results mean — **it does not replace the advice of a qualified doctor.**

■ What might be causing your symptoms?

■ Things to watch out for (Red Flags)

The following were noticed in your results that need prompt attention:

- ****Severe hypoxemia**** (SpO₂ = 85 %)
- ****Leukocytosis**** (WBC = 10,570) suggesting infection/inflammation
- ****Orthostatic dizziness**** indicating possible dehydration or early circulatory compromise
- ****Hyperglycemia & new■onset diabetes**** (fasting glucose = 141 mg/dL, HbA1c = 7.1 %)

■ Tests you may need

Your doctor may ask you to get the following tests done. Each test helps confirm or rule out possible causes:

- ****IV isotonic crystalloid fluids**** (e.g., normal saline 1–2 L bolus, then maintenance) – rehydration.
- ****Oral rehydration solution**** (if patient tolerates oral intake).
- ****Antiemetic**** – ondansetron 4 mg IV/PO q8h PRN.
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■ Possible treatments

These treatments may be considered by your doctor. **Do not take any medication without a doctor's prescription.**

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- ****Oral rehydration solution**** (if patient tolerates oral intake).
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■ Other possibilities the doctor will consider

Medicine isn't always black and white. Here are other conditions that have similar symptoms and may be explored:

- ****Foodborne bacterial gastroenteritis**** (e.g., **Salmonella**, **Campylobacter**, **Staphylococcus aureus**) – recent pizza, acute onset, leukocytosis.
- ****Early sepsis secondary to gastrointestinal infection**** – hypoxemia, leukocytosis, dizziness, possible inadequate perfusion.
- ****Dehydration with orthostatic hypotension**** – dizziness on standing, low SpO₂, poor fluid intake.
- ****Undiagnosed type 2 diabetes mellitus presenting with hyperglycemia and infection**** – elevated fasting glucose and HbA1c.
- ****COVID-19 or other viral respiratory infection presenting atypically**** – low SpO₂ without respiratory complaints warrants exclusion.

■ What should you do next?

1. Share this report with a qualified doctor as soon as possible. The AI has highlighted areas of concern but cannot replace a physical examination.
2. Get the recommended tests done and bring the results to your appointment.
3. If you experience worsening symptoms — especially difficulty breathing, severe pain, or high fever — go to an emergency room immediately.
4. Do not self-medicate based on this report alone.

This plain-language summary is auto-generated for informational purposes only. It is not a substitute for professional medical advice, diagnosis, or treatment. Always seek the advice of a qualified healthcare provider with any questions you may have regarding a medical condition.